AgroSentinel – Pilot Deployment Report (Field Validation)

1. Introduction

This report describes the execution of a real-world pilot of the AgroSentinel MVP, conducted in an olive field located in Maqueda (Toledo, Spain). The objective of this pilot was to validate the integration of multiple data sources—Wireless Sensor Networks (WSN), UAV imagery, GNSS positioning, and Copernicus satellite data—within a unified architecture supported by IPFS and Hyperledger Fabric.

The pilot demonstrates the feasibility of the system in a relevant operational environment, contributing to a TRL 5 / CRL 4 validation.

2. Pilot Overview

The pilot consisted of four main stages:

1. Deployment of WSN nodes on olive trees

2. UAV-based image acquisition with GNSS positioning

3. Retrieval of Copernicus satellite data

4. Data integration into blockchain/IPFS backend

3. WSN Deployment

Two Wireless Sensor Networks were deployed on two separate olive trees. Each WSN consisted of:

- Two sensor nodes (temperature/humidity/environmental pressure ensing)

- One gateway node responsible for data aggregation and transmission

The nodes were physically installed on tree branches to capture localized environmental conditions.

Figure 1a: WSN node installed in tree – wide-view and 1b: WSN node in tree – close-up view

The gateway node transmitted the collected data wirelessly to a nearby laptop, where data was stored locally as structured files. This setup validates:

- Real-world deployability of sensor nodes

- Reliable short-range communication

- Data acquisition under field conditions

4. UAV Data Acquisition with GNSS

A UAV platform was used to capture aerial imagery of the olive field.

A Motorola Edge 50 smartphone compatible with Galileo positioning was attached to a UAV (Parrot ARDrone), enabling:

- Image capture during flight from both devices.

- GNSS-based geolocation of images (UAV and Galileo-enabled smartphone)

Although the device primarily operated using GPS during this test, it is compatible with Galileo, and satellite tracking confirms multi-constellation capability.

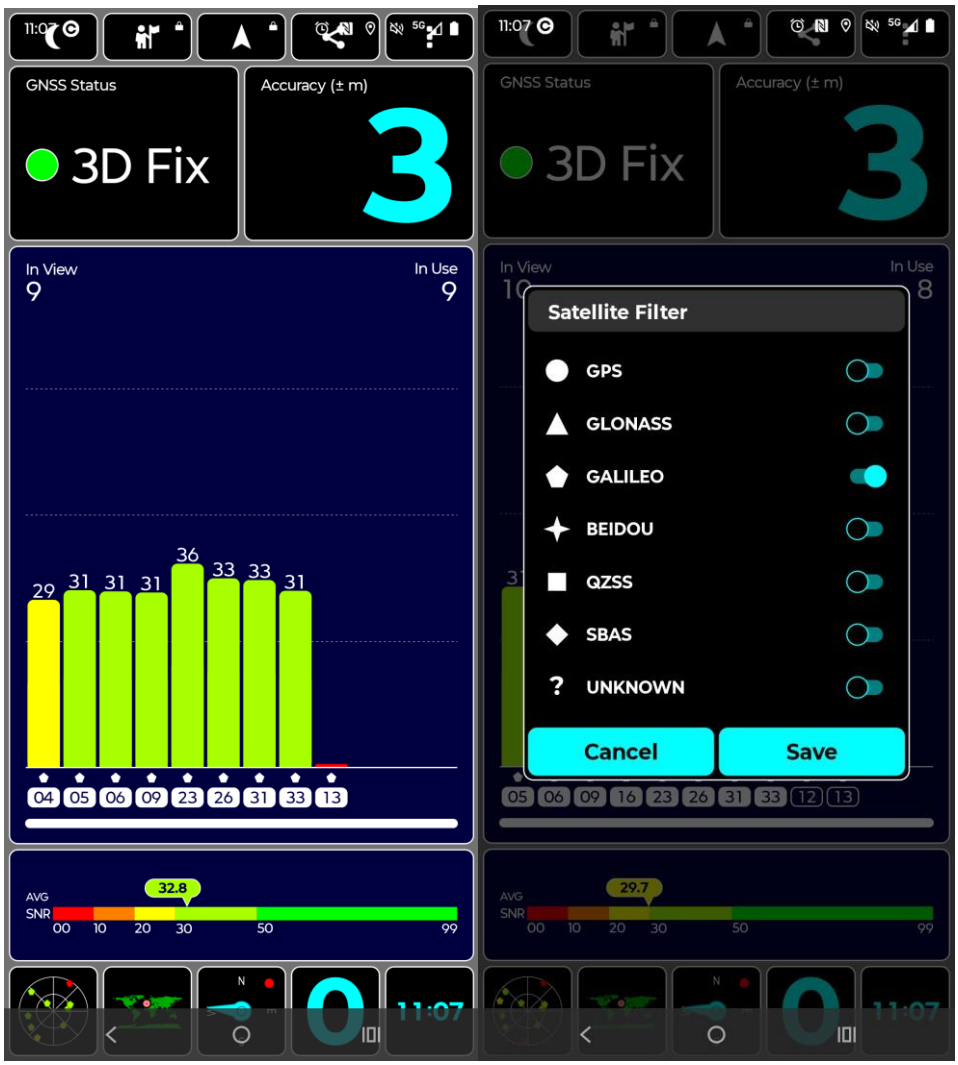

Figure 2a: GNSS status – 3D Fix, satellites in view and 2b: Satellite constellation Galileo-only enabled

Key observations:

- Stable GNSS fix (3D FIX) achieved

- High signal-to-noise ratio (SNR ~30+)

- Multiple satellites in use

This validates:

- Accurate geolocation capability

- Readiness for Galileo-based positioning integration

5. Field Operations

The pilot included active field interaction and system monitoring.

Figure 3: Team operating laptop and WSN in field

Activities included:

- Monitoring incoming WSN data

- Verifying gateway transmission

- Preparing UAV flight

- Collecting environmental observations

This demonstrates:

- End-to-end system operability

- Human-in-the-loop validation

- Field usability of the solution

6. Copernicus Data Integration

Satellite data from the Copernicus programme (Sentinel-2) was retrieved for the same geographic area (Maqueda olive field).

This data provides:

- Large-scale environmental context

- Temporal monitoring capabilities

- Vegetation and crop health indicators

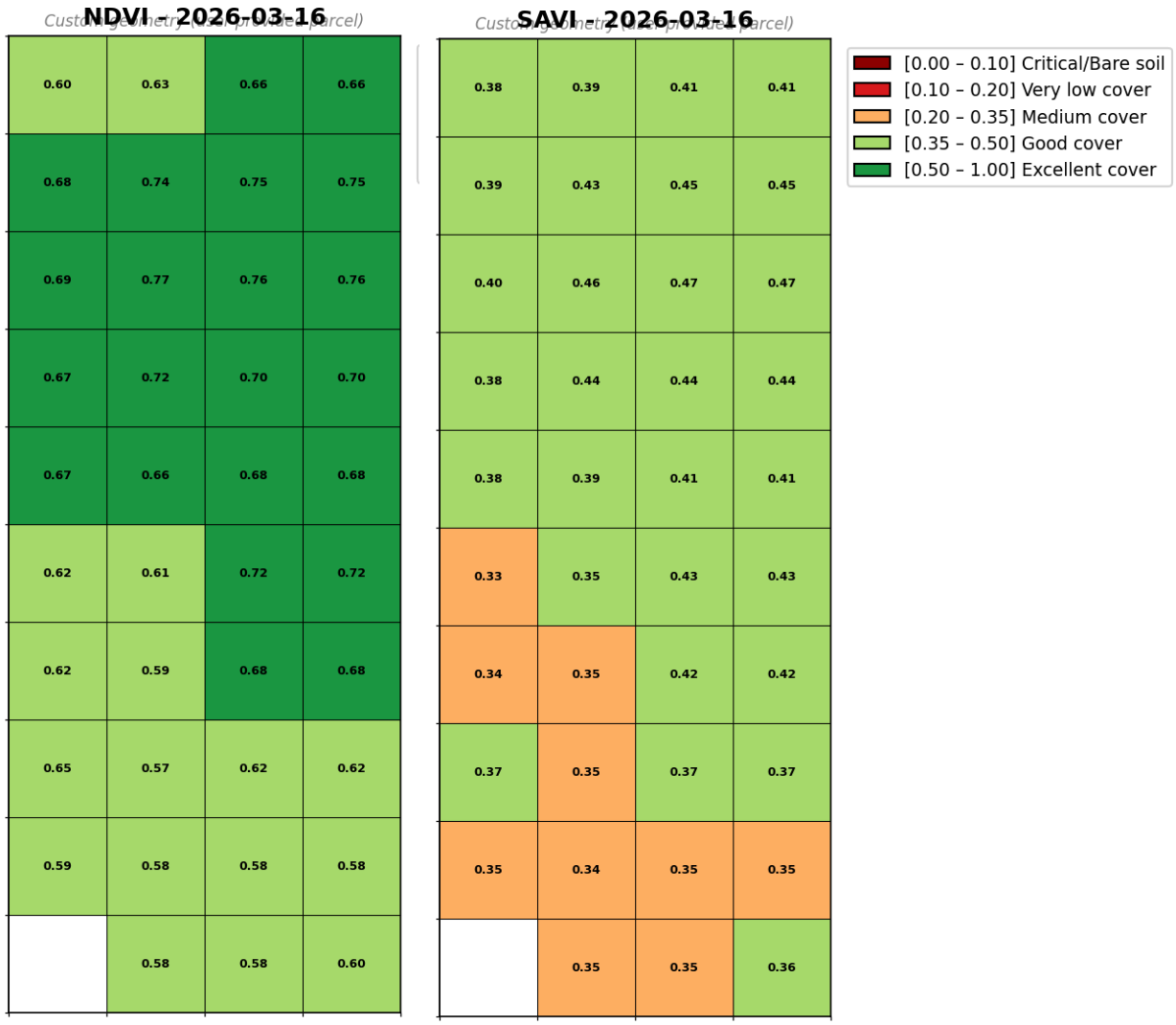

Figure 4: NDVI and SAVI readings obtained for the olive grove by Copernicus

The satellite data complements:

- UAV high-resolution imagery

- WSN localized sensing

7. Data Integration Pipeline

After acquisition, all data sources were integrated into the AgroSentinel backend:

7.1 Data Types Processed

- UAV images

- WSN sensor measurements

- Copernicus satellite data

7.2 Storage and Traceability

- Files stored in IPFS → ensuring decentralized and immutable storage

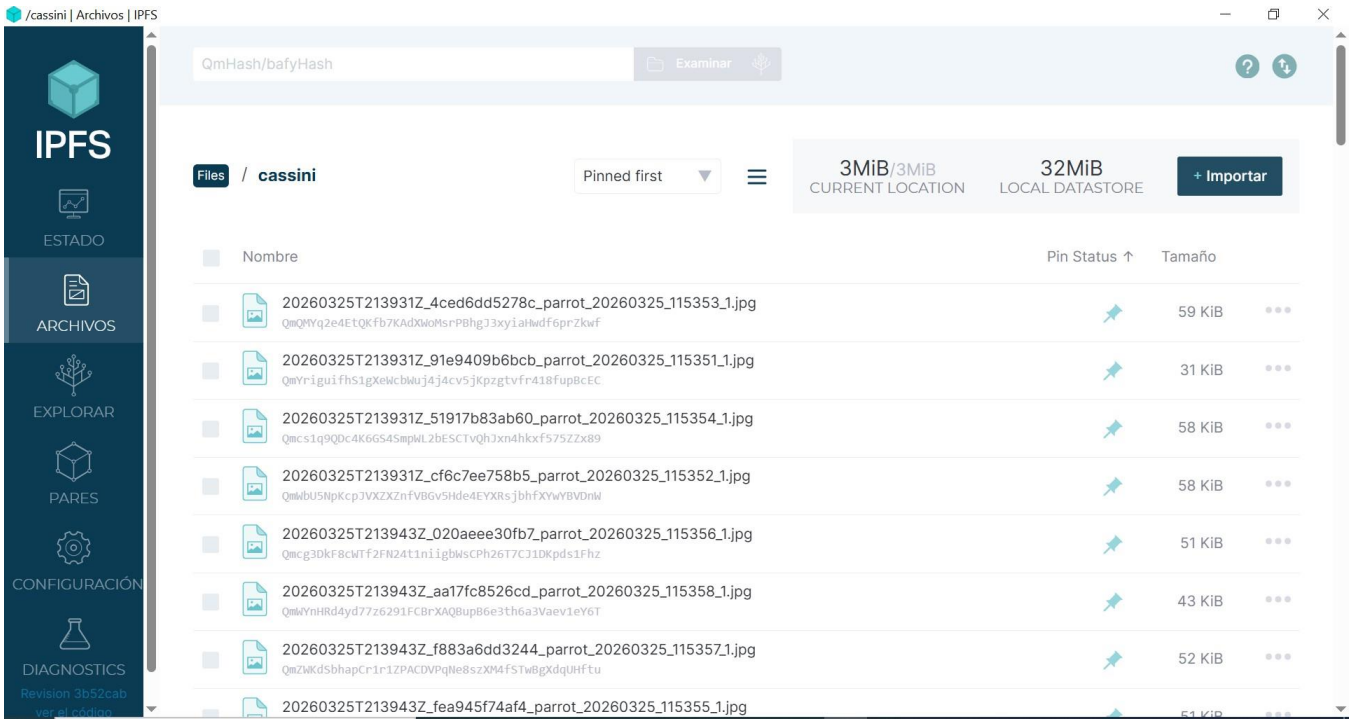

Figure 5: storage of received files in IPFS

- Hashes and metadata recorded in Hyperledger Fabric → ensuring traceability and integrity

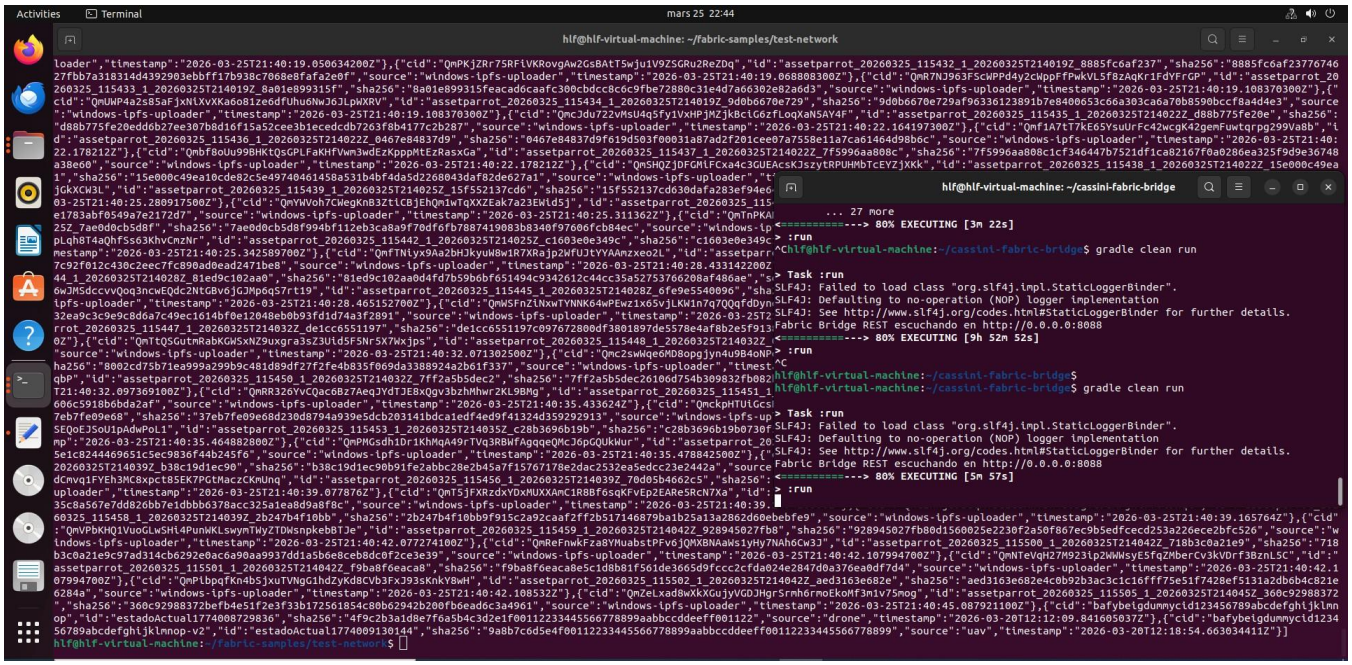

Figure 6: metadata recorded in HyperLedger Fabric linked to the uploaded files in IPFS. Note the REST connection between IPFS and HLF

This enables:

- Verifiable data provenance

- Tamper-proof agricultural records

- Cross-source data linking

8. System Validation Outcomes

The pilot successfully demonstrated:

1. Deployment of WSN in real agricultural environment.
2. UAV-based data acquisition with GNSS positioning.
3. Integration of satellite (Copernicus) data.
4. End-to-end data pipeline (collection → storage → blockchain registration).
5. Multi-source data fusion capability.

9. TRL and CRL Assessment

Based on this pilot: TRL 5 / CRL 4 achieved.

- Technology validated in relevant environment

- Integrated system demonstrated in field conditions

10. Limitations and Next Steps

Identified areas for improvement:

- Full activation and validation of Galileo-only positioning

- Automation of data ingestion pipeline

- Integration of real-time analytics and AI-based crop health assessment

- Development of user dashboard

11. Conclusion

The AgroSentinel pilot confirms the technical feasibility of combining WSN sensing, UAV imagery, GNSS positioning, and Copernicus data within a blockchain-enabled architecture. The system has been successfully validated in a real agricultural setting, strong potential for scalable deployment in precision agriculture and traceability applications.